Exp 5: Develop a customized ping command to Test server connectivity

Aim:

To develop a customized ping command using python to test the connectivity and response time of a remote server.

Procedure:

1) Import socket, struct, os, time and select modules.

2) Create a function to calculate ICMP checksum.

3) Build the ICMP Echo Request packet using struct.pack().

4) Send the packet to the target host using sendto().

5) Calculate and display the round-trip time.

6) Display packet loss percentage.

7) Test with different hosts and record results

OUTPUT:

```
def checksum(data):
    s=0
for i in range(0, len(data),2):
    s+=(data[i] << 8)+data[i+1]
    s=(s>>16)+(s & 0xffff)
    return ~s & 0xffff.

else:
    print('Request timed out for seq=(seq)')

if results:
    print(f'Min=(min(results):.2f)ms Max=(max(
        results):.2f)ms Avg=
    (sum(results)/len(results):.2f)ms')
    sock.close()

ping('google.com').
```

Result:

A customized Ping utility was successfully built. ICMP Echo Requests were sent, replies received, and RTT statistics including minimum, maximum, and average time were displayed.